

A STUDY ON COTPA VIOLATIONS IN KERALA

PROJECT REPORT

Submitted by

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in partial fulfillment of the requirement for the award of
the Degree of Master of Science in Statistics of Kannur
University



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CERTIFICATE

This is to certify that Ms.ATHIRA V, has done the project work entitled "**A STUDY ON COTPA VIOLATIONS IN KERALA**" in partial fulfillment for the award of the MSc Degree in Statistics from Kannur University, under our supervision and guidance .

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DECLARATION

I hereby declare that the Project Report entitled ” **A Study on COTPA violations in Kerala** ” bound herewith is the original record of the project work carried out by me under the supervision and guidance of Dr. Sunila V, Assistant Professor, Department of Statistics, Nehru Arts and Science College, in partial fulfilment of requirements for the award of the Degree of Master of Science in Statistics of University of Kannur and further that no part thereof has been presented before for any other degree.

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Chapter 1

INTRODUCTION

1.1 Introduction

The Cigarettes and Other Tobacco Products Act (COTPA) is a landmark legislation enacted by the Government of India in 2003 to regulate the production, sale, distribution, and consumption of tobacco products. It aims to reduce the demand for tobacco products and protect public health by imposing restrictions on tobacco advertising, promotion, and sponsorship, as well as by mandating health warnings on tobacco product packages. COTPA prohibits smoking in public places, sale of tobacco products to minors, and sale of tobacco products within specified distances from educational institutions. It also mandates the display of health warnings covering a specific percentage of tobacco product packages and prohibits the sale of tobacco products without such warnings.

Additionally, COTPA regulates the packaging and labeling of tobacco products, including requirements for the disclosure of ingredients and emissions. The Act also establishes provisions for the testing and regulation of tobacco products to ensure compliance with specified standards. Violations of this act can result in fines and penalties, and the Act empowers designated officers to enforce its provisions. COTPA has played a crucial role in raising awareness about the health risks associated with tobacco use and in promoting tobacco control measures in India.

COTPA (Cigarettes and Other Tobacco Products Act) violations in Kerala, like in many other places, can occur due to various reasons. Some common reasons include:- Lack of Awareness:

Many people may not be fully aware of the provisions of COTPA and the specific regulations regarding tobacco use and sale, Ease of Availability: Despite regulations, tobacco products are often readily available, which can lead to increased use and violations, Addiction: Nicotine addiction can drive individuals to violate COTPA regulations, such as smoking in prohibited areas, Social Norms: In some communities, smoking or using tobacco products may be considered socially acceptable, leading to less compliance with COTPA regulations, Lack of Alternatives: Some individuals may not have access to or knowledge about alternative products or methods to quit tobacco, leading them to continue using tobacco products in violation, Influence of Media: Portrayal of smoking or tobacco use in movies, TV shows, or other media can influence behavior and contribute to violations etc.

COTPA violations are important to address because they indicate non-compliance with the law, which can lead to increased tobacco consumption and related health issues. By enforcing COTPA, authorities can reduce the availability of tobacco products to minors, protect non-smokers from exposure to tobacco smoke, and promote public health. This Act includes various sections that outline prohibitions and penalties for violations :

- Section 4 : Prohibition of smoking in public places.
- Section 5 : Prohibition of advertisement of cigarettes and other tobacco products
- Section 6 : Prohibition of sale of cigarettes and other tobacco products to minors.
- Sub-section 6 (a): Prohibition of sale of tobacco products within a specified area around educational institutions.
- Sub-section 6 (b): Prohibition of sale of tobacco products through the internet or any other electronic media.
- Section 7 : Prohibition on trade and commerce in production, supply and distribution of cigarettes and other tobacco products.

The Act also gives power to any police officer, not below the rank of a sub-inspector or any officer of State Food or Drug Administration

or any other officer, holding the equivalent rank being not below the rank of Sub-Inspector of police for search and seizure of premises where tobacco products are produced, stored or sold, if he suspects that the provision of the Act has been violated.

A person who manufactures tobacco products and fails to adhere to the norm related to warnings on packages on first conviction shall be punished with up to 2 years in imprisonment or with fine which can extend to Rs. 5000, in case of subsequent conviction shall be punished with up to 5 years in imprisonment or with fine which can extend to Rs. 10000 and a fine up to Rs. 200 can be imposed for smoking in public place, selling tobacco products to minors, or selling tobacco products within a radius of 100 metres from any educational institution. A person who advertises tobacco products shall on first conviction shall be punished with up to 2 years in imprisonment or with fine which can extend to Rs. 1000, in case of subsequent conviction shall be punished with up to 5 years in imprisonment or with fine which can extend to Rs. 5000. The Act repealed The Cigarettes (Regulation of Production, Supply and Distribution) Act, 1975.

1.1.1 Objectives

The purpose of this study is to analyze violations of the Cigarettes and Other Tobacco Products Act (COTPA) in Kerala.

- Identifying the most violated section of the act.
- Forecasting future violation of the most violated act.

1.1.2 About Data and Study

This study forecasting most violated section of COTPA in Kerala, is crucial for enhancing enforcement efficiency and public health protection. Kerala, known for its progressive stance on public health issues, has been actively enforcing COTPA, yet violations persist, presenting significant challenges.

Data for the study is collected from the official website of Kerala police department(www.keralapolice.gov.in). Study is based on the latest available data from January 2019 to December 2023.

The enforcement of the Cigarettes and Other Tobacco Products Act (COTPA) in Kerala has led to significant numbers of violations being recorded over the years. According to the latest data from the Kerala Police, COTPA violations are mainly categorized under different sections, with Section 4 being the most frequently violated. This section, which pertains to the prohibition of smoking in public places, saw tens of thousands of violations annually in recent years. For instance, in 2023 alone, there were over 76,000 violations under Section 4, and by early 2024, nearly 10,000 violations had already been reported.

Other sections, such as Section 6(b) (prohibiting the sale of tobacco products to minors), also see a significant number of violations but on a smaller scale compared to Section 4. The total number of violations and the fines collected illustrate the ongoing challenge of enforcing tobacco control laws in the state.

From the forecasted values of most violated section (section4) we can understand how much continues monitoring needed for reducing future violation of COTPA aiming to reduce the health risks associated with tobacco use in Kerala.

Chapter 2

METHODOLOGY

2.1 Methodology

First and foremost step in analysis is study the data in details. Some important terms that are used in time series analysis is as follows.

2.1.1 Time Series

Time series analysis is a statistical technique used to analyze and interpret time-ordered data. It involves studying the pattern of data points collected at successive and equally spaced intervals of time. Time series analysis can help in understanding the underlying structure of the data, making forecasts, and drawing meaningful insights for decision-making.

The purpose of time series analysis is generally twofold: to understand or model the stochastic mechanisms that gives rise to an observed series and to predict or forecast the future values of a series based on the history of that series. The skill of a time series forecasting model is determined by its performance at predicting the future. This is often at the expense of being able to explain why a specific prediction was made, confidence intervals and even better understanding the underlying causes behind the problem.

Utility of Time Series Analysis

The analysis of Time Series is of great significance not only to the economist and businessman but also to the scientist, geologist, biologist, research worker, etc...

- It helps in understanding past behaviors: By observing data over a period of time one can easily understand what changes have taken place in the past, Such analysis will be extremely helpful in producing future behavior.
- It helps in planning future operations: Plans for the future cannot be made without forecasting events and relationship they will have. On the basis of the past and present condition future is well predicted.
- It helps in evaluating current accomplishments: by providing a structured way to examine historical data and assess performance over time. Eg; Performance Tracking- A company can use time series analysis to evaluate sales performance over different periods (monthly, quarterly, annually) to see if sales targets are being met and to understand seasonal variations.
- It facilitates comparison: Different time series are often compared and important conclusions drawn there from.

Components of Time Series

Time series data can be decomposed into several key components, each of which helps to understand the underlying patterns and structure of the data. These components are:

- **Trend:** The long-term movement in the data that shows the general direction (upward or downward) over a long period. This component represents the underlying growth or decline in the series.
- **Seasonality:** Regular, periodic fluctuations that occur at fixed intervals (such as daily, monthly, quarterly, or yearly). Seasonal patterns repeat over a specific period and are often driven by factors like weather, holidays, or other recurring events.
- **Cyclical Patterns:** Fluctuations that occur over irregular intervals, often related to economic or business cycles. These cycles are not fixed in length and can span several years. Unlike seasonality, cyclical patterns do not have a regular, predictable period.

- Irregular Component (Noise): Random variations or residuals that cannot be attributed to trend, seasonality, or cyclical patterns. This component represents the unpredictable, random noise in the data and can be caused by one-time events or random fluctuations.

Decomposing a time series into these components helps in better understanding and analyzing the data.

The following are the two models which we generally use for the decomposition of time series into its four components. The objective is to estimate and separate the four types of variations and to bring out the relative effect of each on the overall behavior of the time series.

The decomposition can be either additive or multiplicative. The choice between these models depends on the nature of the time series data.

1. Additive model
2. Multiplicative model

- In additive model, a time series can be expressed as,

$$Y_t = T_t + S_t + C_t + R_t$$

Where Y_t is the time series values at time 't', T_t , S_t , C_t and R_t represent the trend, seasonal, cyclical and random fluctuations at time 't'. The additive model assumes that components operate independently and can be linearly combined, which is particularly useful when the amplitude of seasonal variations does not change with the level of the time series. S_t won't appear in a series of annual data.

The additive model is useful when the seasonal variation is relatively constant over time.

- In multiplicative model, a time series can be expressed as,

$$Y_t = T_t \cdot S_t \cdot C_t \cdot R_t$$

$$\log Y_t = \log T_t + \log S_t + \log C_t + \log R_t$$

This is the converted multiplicative model into additive model by taking logarithms.

A multiplicative model in time series analysis is used when seasonal variations or trends change proportionally to the level of the time series. This model is particularly useful for data that exhibit increasing or decreasing seasonal fluctuations over time.

While forecasting a given time series data we need to find out appropriate models to get accurate results.

Definition:

If a random variable X is indexed to time, usually by t , the observations $\{X_t, t \in T\}$ is called a time series, where T is a time series index set. An intrinsic feature of a time series is that, typically adjacent observations are independent. A time series is said to be continuous when the observations are made continuously on time. Time series data is said to be discrete when the observations are taken only at specific times, usually equally spaced.

A give time series can be,

- Stationary
- Non Stationary

2.1.2 Stationary

A stationary time series is one whose statistical properties such as mean, variance, and autocorrelation structure do not change over time. In simpler terms, it has constant statistical properties over time. Many statistical methods and models, including ARIMA (AutoRegressive Integrated Moving Average), assume that the time series data is stationary. Therefore, testing for stationarity is an essential step before model fitting.

Weakly stationary

A stochastic random variable $\{X_t, t \in T\}$ is said to be weakly stationary, if it satisfies the following properties.

- The mean $E(X_t)$ is same for all t .
- The variance of X_t is same for all t .

- The covariance (and also correlation) between X_t and X_{t-h} is same for all t, h .

Strictly stationary

A stationary process $\{X_t, t \in T\}$ is strictly stationary, its probability distribution is invaried under the translation of time axis.

ie, for time $t_1, t_2, \dots, t_n$ the joint distribution of $(X_{t_1}, X_{t_2}, \dots, X_{t_n})$ is same as the distribution of $(X_{t_1+h}, X_{t_2+h}, \dots, X_{t_n+h})$.

2.1.3 Auto Regressive (AR) Process

An autoregressive (AR) process is a type of stochastic process commonly used in time series analysis to model the behavior of a variable over time. In an AR process, each value of the time series is a linear combination of its own past values and a random error term. In other words, the value of the variable at time t is predicted based on its past values, with the weights or coefficients of the past values being estimated from the data.

Mathematically, an AR process of order p can be denoted as $AR(p)$, where “ p ” is called the order of the model and represents the number of lagged values we want to include.

A process considered $AR(1)$ is the first order process. It means that the current value based on the immediately preceding value. An $AR(2)$ process has the current value based on the previous two values.

$AR(p)$ model has the form,

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + Z_t$$

Here $\phi_1, \phi_2, \dots, \phi_p$ are used to represent finite set of weight parameter.

We can rewrite $AR(P)$ model in terms of backward difference operator B as,

$$X_t = \phi_1 B X_t + \phi_2 B^2 X_t + \dots + \phi_P B^P X_t + Z_t$$

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_P B^P) X_t = Z_t$$

OR

$$X_t = \frac{1}{\phi(B)} Z_t$$

2.1.4 Moving Average (MA) Process

In time series analysis, the moving-average model (MA model), also known as moving-average process, is a common approach for modeling univariate time series. The moving-average model specifies that the output variable depends linearly on the current and various past values of a stochastic term. MA process with order "q" is denoted by MA(q), order "q" represents the number of past error terms that included in the model.

MA(q) model defined as,

$$X_t = Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \dots - \theta_q Z_{t-q}$$

2.1.5 Autoregressive-Moving average (ARMA) process

An ARMA (AutoRegressive Moving Average) process is a model used in time series analysis to describe a weakly stationary stochastic process. It combines two components: autoregression and moving average. The ARMA model is used to understand and predict future points in a time series based on its own past values and past errors. If the process has terms from both an AR(p) and MA(q) process, then the process is called ARMA(p, q) and can be expressed as,

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + Z_t - \sum_{j=1}^q \theta_j Z_{t-j}$$

OR

$$\phi(B)X_t = \theta(B)Z_t$$

This is called mixed autoregressive-moving average process of order (p,q) and is abbreviated as ARMA(p,q).

2.1.6 Differencing

Differencing is a crucial technique in time series analysis used to make a series stationary. Stationarity is important because many statistical methods and models particularly autoregressive integrated moving average (ARIMA) models. assume that the properties of the

series do not change over time. Understanding when and how to apply differencing, as well as the implications of over-differencing, is crucial for accurate and effective time series analysis.

- First-Order Differencing

First-order differencing is the process of subtracting the current observation from the previous one:

$$y'_t = y_t - y_{t-1}$$

This is useful for removing linear trends in the data. If the original series y_t is non-stationary, the differenced series y'_t might be stationary.

The differenced series will have only $n - 1$ values, since it is not possible to calculate a difference y'_1 for the first observation.

- Second-Order Differencing

Second-order differencing is used when first-order differencing is not enough to achieve stationarity. It involves differencing the first-order differenced series: The second-order differencing of the time series is given by:

$$y''_t = y'_t - y'_{t-1} = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) = y_t - 2y_{t-1} + y_{t-2}$$

In this case y''_t will have $n-2$ values. In practice, it is almost never necessary to go beyond second-order differences.

- Seasonal Differencing

A seasonal difference is the difference between an observation and the previous observation from the same season. So

$$y'_t = y_t - y_{t-m}$$

where m = the number of seasons. This type of differencing is used to remove seasonal effects from the series. Proper identification and application of seasonal differencing can significantly enhance the performance and interpretability of time series models.

2.1.7 Auto Regressive Integrated Moving Average (ARIMA) Model

An autoregressive integrated moving average (ARIMA) model is a generalization of an autoregressive moving average (ARMA) model. To better comprehend the data or to forecast upcoming series points, ARIMA models are applied in some cases where data show evidence of non-stationarity in the sense of mean (but not variance/autocovariance), where an initial differencing step (corresponding to the "integrated" part of the model) can be applied one or more times to eliminate the non-stationarity of the mean function. When the seasonality shows in a time series, the seasonal-differencing could be applied to eliminate the seasonal component.

Non-seasonal ARIMA models are generally denoted ARIMA(p,d,q) where parameters p, d, and q are non-negative integers, p is the order (number of time lags) of the autoregressive model, d is the degree of differencing, and q is the order of the moving-average model. Seasonal ARIMA models are usually denoted ARIMA(p,d,q)(P,D,Q)m, where m refers to the number of periods in each season, and the uppercase P,D,Q refer to the autoregressive, differencing, and moving average terms for the seasonal part of the ARIMA model.

A general ARIMA model can be written as:

$$B_t^d = \frac{\theta(B)Z_t}{\phi(B)}$$

Where Z_t is assumed to have white noise, B is the backward difference operator and d is the order regular differencing needed to achieve time series stationary.

Other parameters are defined as:

$$\theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q$$

$$\phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$$

Where, $\phi_1, \phi_2, \dots, \phi_p$ are AR parameters and $\theta_1, \theta_2, \dots, \theta_q$ are MA parameters.

2.1.8 Auto Correlation Function

The autocorrelation function (ACF) is a statistical tool used to measure the correlation between observations of a time series separated by different time lags. It helps identify the extent to which current values of the series are related to past values.

The autocorrelation function (ACF) at lag k , denoted as $\rho(k)$, is given by:

$$\rho(k) = \frac{\sum_{t=1}^{n-k} (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2} = \frac{cov(x_t, x_{t-k})}{var(x_t)} = \frac{\gamma(k)}{\gamma(0)}$$

A plot of the ACF against the lag is known as the correlogram. It is frequently used to identify whether or not seasonality is present in a given time series and also to determine if data are stationary.

2.1.9 Partial Auto Correlation Function(PACF)

In time series analysis, the partial autocorrelation function (PACF) gives the partial correlation of a stationary time series with its own lagged values, regressed the values of the time series at all shorter lags.

Unlike the autocorrelation function (ACF), which measures the correlation between two observations in a time series at different lags, the PACF measures the correlation between two observations at a particular lag, controlling for the effects of the other observations in between. The PACF often used in time series analysis and modelling to identify the order of an autoregressive (AR) process. The k^{th} order PACF is defined as the coefficient δ_k in the following regression model,

$$X_t = \delta_0 + \delta_1 X_{t-1} + \delta_2 X_{t-2} + \dots + \delta_k X_{t-k} + \epsilon_t$$

PACF(k) measures the "direct" impact of X_{t-k} on X_t obviously X_{t-k} also intermediately affect X_t through $X_{t-k+1} \dots X_{t-1}$.

2.1.10 Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller (ADF) test is a statistical test used to determine whether a given time series is stationary or contains a unit root, implying non-stationarity. Stationarity is a crucial property in time series analysis, as many statistical models assume that the underlying time series is stationary. A stationary time series has constant statistical properties, such as mean and variance, over time. Non-stationary time series, on the other hand, exhibit trends, cycles, or other structures that change over time.

2.1.11 Forecasting

Forecasting is a crucial analytical technique used to predict future values based on historical data. It plays a vital role in decision-making across various fields such as finance, economics, weather prediction, and more. By analyzing patterns and trends within a dataset, forecasting models can provide insights into future occurrences, enabling proactive planning and strategic actions. Common methods include time series analysis, such as ARIMA (AutoRegressive Integrated Moving Average), exponential smoothing, and machine learning approaches. Accurate forecasting hinges on data quality, appropriate model selection, and thorough validation, ensuring reliable predictions that can guide both short-term and long-term decisions. In this study we used ARIMA model.

2.1.12 Residual Analysis

After fitting the forecasting model, we conducted a residual analysis to ensure the model's adequacy. It helps in assessing the goodness-of-fit of the model and ensuring that the model assumptions are met. The residuals, which are the differences between the observed values and the predicted values.

$$\text{Residual} = \text{Observations} - \text{fitted values}$$

To check whether the residuals are uncorrelated, we calculate their sample autocorrelation and partial autocorrelation.

2.1.13 Model Adequacy

Model adequacy refers to the extent to which a statistical or predictive model accurately represents the real-world data it aims to describe or predict. Assessing model adequacy involves evaluating whether the assumptions underpinning the model are valid and if the model's predictions align well with actual observations. This process includes diagnostic checks such as residual analysis, goodness-of-fit tests, and validation against independent data sets. A model is deemed adequate if it demonstrates robustness, minimal bias, and reasonable predictive accuracy. Ensuring model adequacy is crucial for the model's reliability and the validity of any inferences or decisions derived from it.

To check the model is adequate, the following procedure should be adopted.

2.1.14 Ljung-Box Test

The Ljung-Box test, also known as the Ljung-Box Q test, is a statistical test used to determine if there is any significant autocorrelation in a time series data set. This test helps to assess whether the residuals from a time series model, like an ARIMA model, are independently distributed or if there are patterns that suggest otherwise.

The null and alternative hypothesis of the test as follows,

H_0 : The data are independently distributed; there is no autocorrelation up to a specified lag k .

H_1 : The data are not independently distributed; there is significant autocorrelation up to a specified lag k .

The test statistic is:

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$$

Where;

- n is the number of observations.
- h is the number of lags being tested.

- $\hat{\rho}_k^2$ is the sample autocorrelation at lag k.

Under H_0 : the statistic Q asymptotically follows a $\chi^2(h)$. For significance level α , the critical region for rejection of the hypothesis of randomness is:

$$Q > \chi_{1-\alpha, h}^2$$

Where $\chi_{1-\alpha, h}^2$ is the $(1 - \alpha)$ quantile of the chi-squared distribution with h degrees of freedom.

2.1.15 Shapiro Wilk Test

The Shapiro-Wilk test is a statistical test used to assess whether a given sample comes from a normally distributed population. The test statistic is,

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)}\right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Where:

- $x_{(i)}$ are the ordered sample values.
- $\bar{x}$ is the sample mean.
- a_i are constants generated from the means, variances, and covariances of the order statistics of a sample from a standard normal distribution.

The null hypothesis of the test is that the population is normally distributed.

- If the p-value is greater than the chosen alpha level we fail to reject the null hypothesis, suggesting that the data does not significantly deviate from normality.
- If the p-value is less than the chosen alpha level, we reject the null hypothesis, indicating that the data significantly deviates from normality.

2.1.16 Akaike Information Criterion(AIC)

The Akaike information criterion(AIC) is a measure of the relative quality of a statistical model, for a given data. It was developed by Hirotugu Akaike and has since become a widely used tool in various fields, including statistics, econometrics, and machine learning. The AIC provides a way to assess the relative quality of different models by balancing the goodness of fit and the complexity of the model. It aims to find the model that achieves a good fit to the data while using the fewest parameters possible. The principle underlying AIC is based on information theory and involves quantifying the trade-off between model complexity and predictive performance. The computation of AIC involves two components: the goodness of fit and a penalty term for model complexity. The goodness of fit is typically measured using the maximized log-likelihood of the model, which represents how well the model fits the observed data. The penalty term is proportional to the number of parameters in the model.

The AIC value for a specific model is obtained by subtracting twice the log-likelihood from twice the number of parameters.

$$AIC = 2k - 2\ln(L)$$

- A lower AIC value indicates a better model.
- AIC can only be used to compare models that are fitted to the same dataset. It does not provide an absolute measure of model quality but a relative one.

2.1.17 Bayesian Information Criterion (BIC)

Bayesian information criterion (BIC) or Schwarz information criterion (SIC) is a statistical criterion for model selection among a finite set of models. It is based on the likelihood function and is penalized for the number of parameters in the model to discourage overfitting. In the context of time series analysis, BIC is particularly useful for comparing different time series models to identify the one that best balances goodness of fit and model complexity.

It is based, in part, on the likelihood function and it is closely related to the Akaike information criterion (AIC). When fitting models, it is possible to increase the maximum likelihood by adding parameters, but doing so may result in overfitting. Both BIC and AIC attempt to resolve this problem by introducing a penalty term for the number of parameters in the model; the penalty term is larger in BIC than in AIC for sample sizes greater than 7.

The BIC is given by the formula:

$$BIC = k \ln(n) - 2 \ln(\hat{L})$$

where:

- $(\hat{L})$ is the maximized value of the likelihood function of the model
- k is the number of parameters in the model.
- n is the number of observations.

When picking from several models, ones with lower BIC values are generally preferred.

2.1.18 Q-Q Plot

A Q-Q plot, or quantile-quantile plot, is a graphical tool for comparing two sets of data to assess whether they come from the same distribution. In the context of time series modeling, a Q-Q plot can be used to check whether the residuals of a fitted time series model follow the normal distribution. If the data points in the plot closely follow a straight line, it indicates that the dataset is approximately normally distributed. Deviations from the line suggest departures from normality, which may require further investigation or non-parametric statistical techniques.

2.1.19 Forecasting procedure

We prepare time series plot for the number of violations section 4 of COTPA and then decompose the observed data into trend, seasonal and random components. Next we prepare the seasonal plot, ACF and PACF of the data. By analyzing the ACF and PACF plots, we can determine the appropriate values for p and q for your ARIMA model. The ACF helps identify the moving average component q , while the PACF helps identify the autoregressive component p .

2.1.20 ARIMA Model Fitting and Forecasting

First we want to check stationarity of the data. Here Augmented Dickey-Fuller (ADF) Test is used for checking stationarity. If not stationary the data is differenced and further checked stationarity. Differencing was performed in step 1 to achieve stationarity, the differencing order (d) is already determined. The number of differences needed corresponds to the value d . After identifying the values of p , d and q , specify the ARIMA model as $ARIMA(p, d, q)$. Further we plot residuals for checking correlation and test the randomness by Ljung-Box Test. If the values are uncorrelated the forecasts are good.

Chapter 3

ANALYSIS

3.1 Analysis

3.1.1 Introduction

In this section, we performed an in-depth time series analysis on the dataset spanning from 2019 to 2023. Initially visualized the data using barplot to identify which section is most violated. After that we visualized the most violated section data using line graphs to identify underlying trends and seasonal patterns. These graphs revealed trend and distinct seasonal variations. To quantify these patterns, we employed the Seasonal Autoregressive Integrated Moving Average (ARIMA) model, which was selected based on its suitability for our data characteristics. The ARIMA model parameters were determined using the ACF and PACF plots. The model was then used to forecast future values, as illustrated in the forecast plot.

3.1.2 Barplot of COTPA section violations

From Figure 1, understood that the most violated section of COTPA is Section 4. Violations of Section 5, Section 6(a), section 6(b), Section 7 are very low compared to Section 4.

Studying most violated section (Section 4) of COTPA violations.

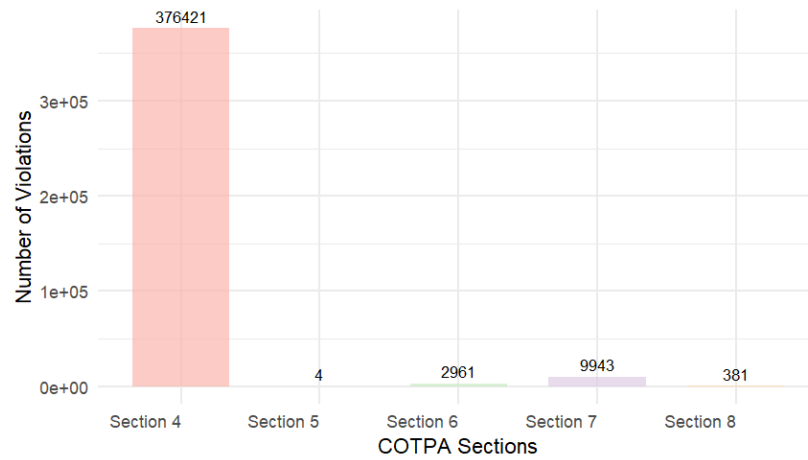


Figure 3.1: Violations of COTPA sections of 5 years.

3.1.3 Time Series Plot

A time series plot is the primary step in time series analysis, providing a visual representation of data points collected or recorded at specific time intervals. This plot displays the data along the y-axis against time on the x-axis.

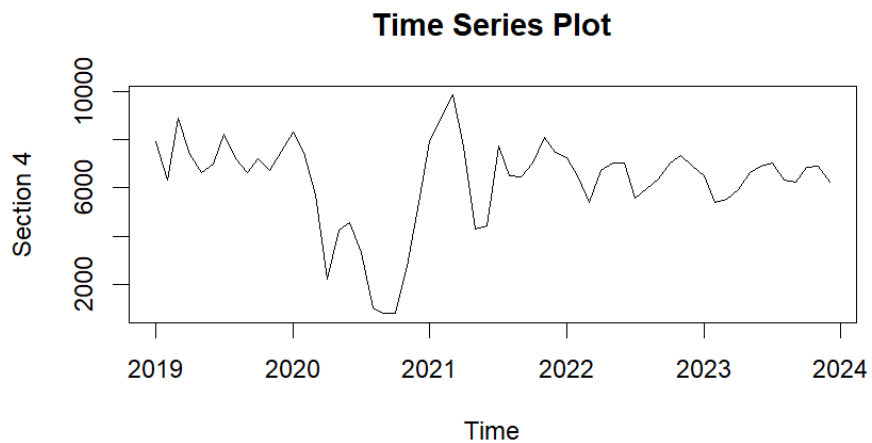


Figure 3.2: Time Series Plot of section4 COTPA Violation.

3.1.4 ACF and PACF Plots

After plotting the time series, the next critical step in time series analysis is to examine the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots. These plots are essential to check for stationarity of the data and for identifying suitable models for forecasting.

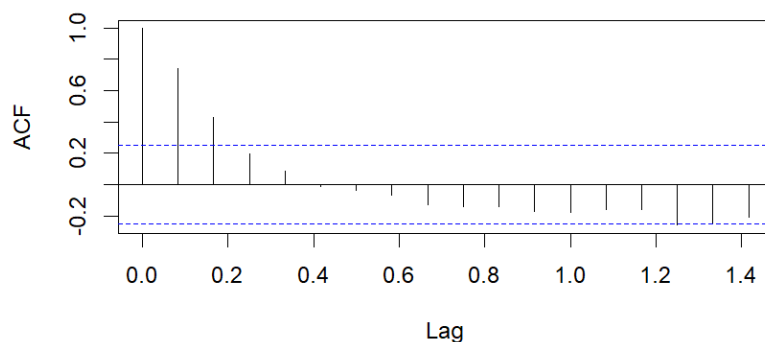


Figure 3.3: ACF Plot.

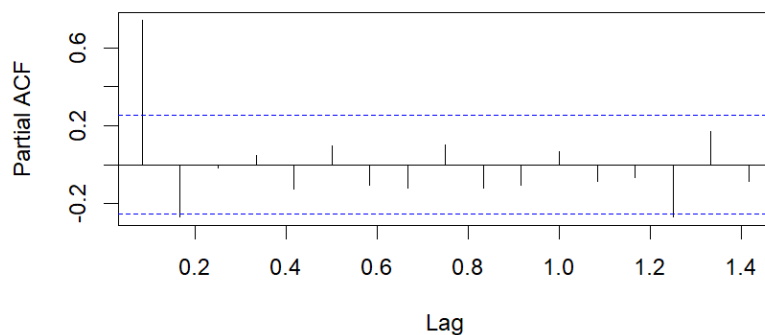


Figure 3.4: PACF Plot.

ACF and PACF plots shows the data is non stationary.

Augmented-Dickey Fuller Test

The stationarity of time series data can be tested using Augmented-Dickey Fuller test. Here we test the hypothesis,

H_0 : COTPA Section4 violation data is non-stationary

Against

H_1 : COTPA Section4 violation data is stationary

The result of the test is,

Augmented-Dickey Fuller Test	Lag parameter	p-value
-2.7382	3	0.2764

Here p value is greater than 0.05, so we accept the null hypothesis that the data is non-stationary.

3.1.5 Decomposition of Time Series Data

Perform decomposition of the time series data for evaluating the trend, seasonal and random components.

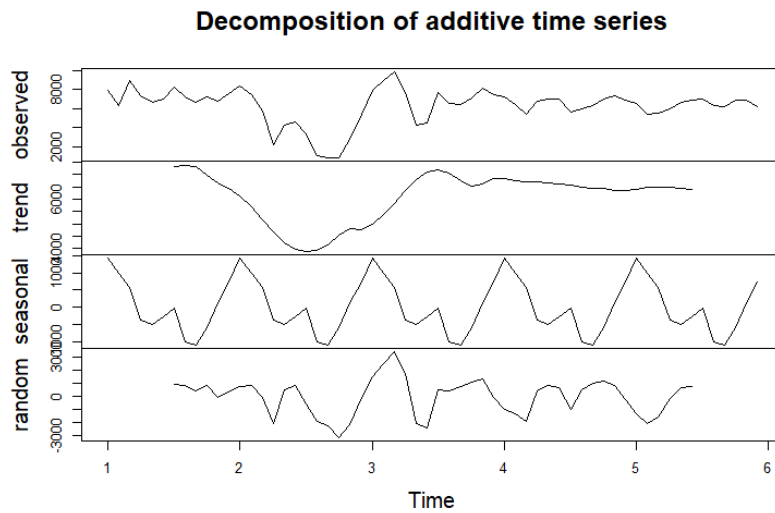


Figure 3.5: Decomposed Data

From Figure:5 The second plot represents the trend component, indicating a generally a small downward trend from 2019 to early 2021, followed by a relatively stable period until 2024.

The third plot illustrates the seasonal component, showing regular, repeating patterns. The seasonality appears to be strong and periodic, with peaks and troughs occurring at regular intervals. This confirms the presence of seasonality in the time series.

The bottom plot represents the random or residual component, showing the irregular fluctuations that are not explained by the trend or seasonal components. These residuals appear to be relatively stable, suggesting that the model has effectively captured the major patterns in the data.

Differencing is a common technique used in time series analysis to remove trends and seasonality, thereby transforming a non-stationary series into a stationary one. So we difference the data,

3.1.6 Time Series Plot of Differenced Series

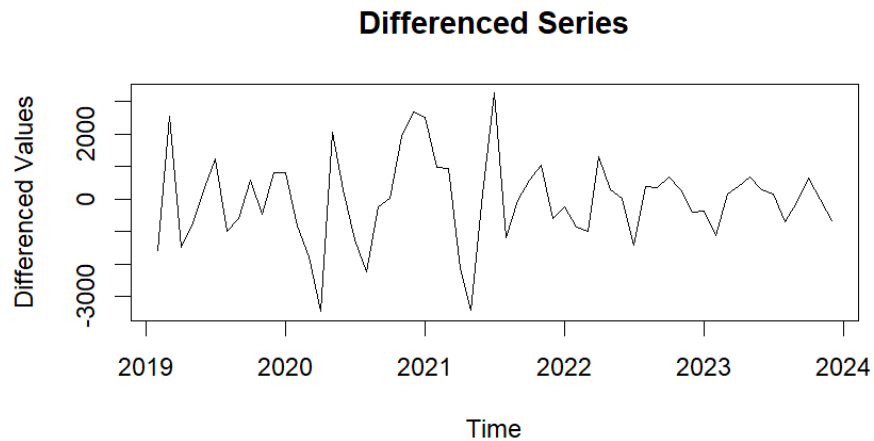


Figure 3.6: Plot after differencing

3.1.7 ACF and PACF Plots

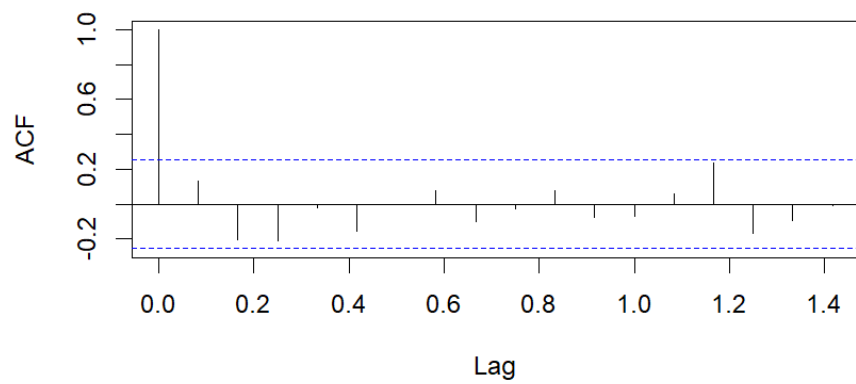


Figure 3.7: ACF Plot after differencing

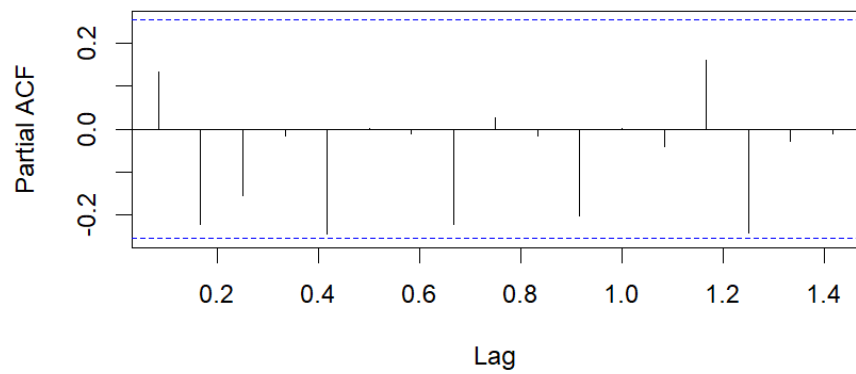


Figure 3.8: PACF Plot after differencing.

Now we again check stationarity using ADF test.

Here we test the hypothesis,

H_0 : COTPA Section4 violation data is non-stationary

Against

H_1 : COTPA Section4 violation data is stationary

The result of the test is,

Augmented-Dickey Fuller Test	Lag parameter	p-value
-4.1961	3	0.01

Here the test gives p value 0.01 which is less than 0.05, so we reject the null hypothesis and hence conclude that the data is stationary.

3.1.8 ARIMA model

Here we choose the best model from all the possible models according to Akaike Information Criterion. The best model is ARIMA(2, 1, 1)(1, 0, 0)_[12] with lowest AIC = 1015.08 and BIC = 1025.4

The model coefficients are give by,

	ar1	ar2	ma1	sar1
	0.9716	-0.2821	-1.0	-0.0789
s.e.	0.1256	0.1253	0.0558	0.1339

3.1.9 Residual Analysis

Residual analysis is used to check the model adequacy. i.e, the forecast errors of our model are normally distributed , whethere there is any correlation between successive forecast errors.

The time series plot of residuals are given by,

3.1.10 Residual plot

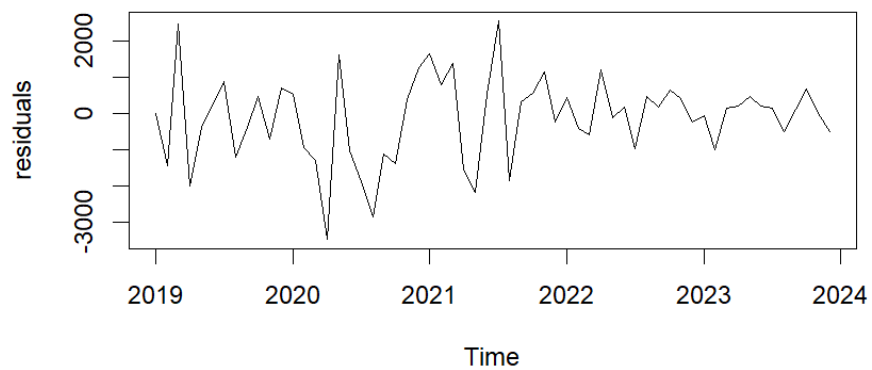


Figure 3.9: Time series plot of residuals

3.1.11 ACF and PACF plot of Residual

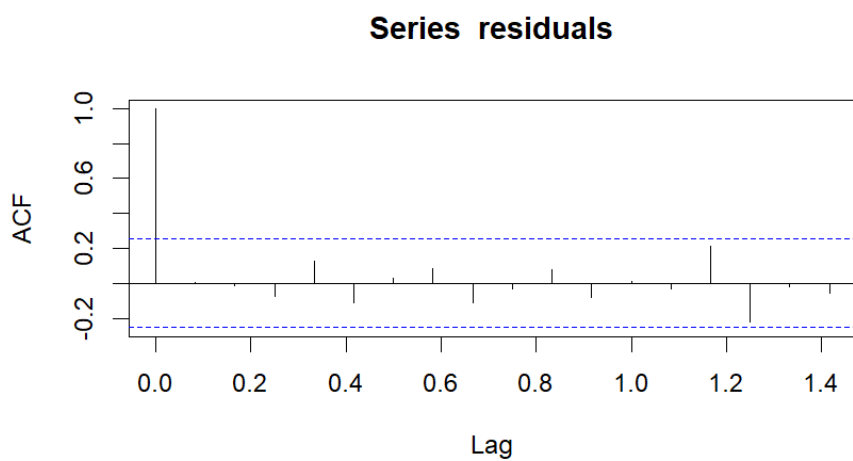


Figure 3.10: Acf plot of residuals

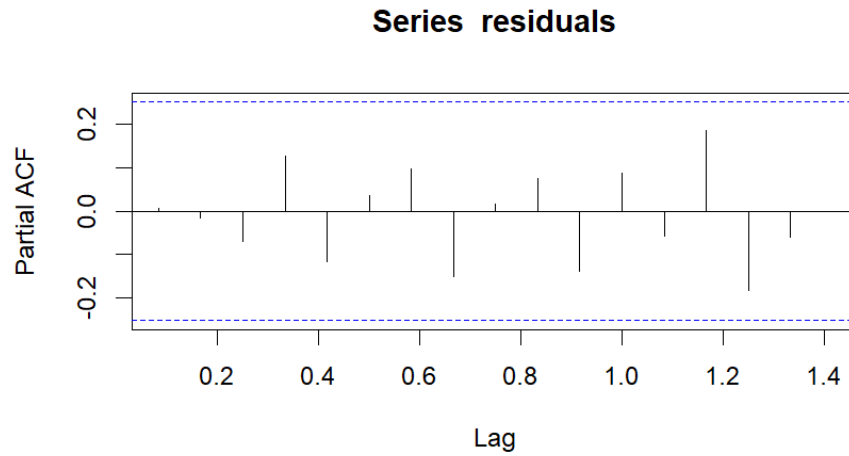


Figure 3.11: PAcf plot of residuals

The ACF and PACF plots of the residuals from the fitted model show that all spikes are within the significant limits. This indicates that the residuals do not exhibit significant autocorrelation at any lag, suggesting that the residuals are approximately white noise.

3.1.12 Q-Q Plot of Residuals

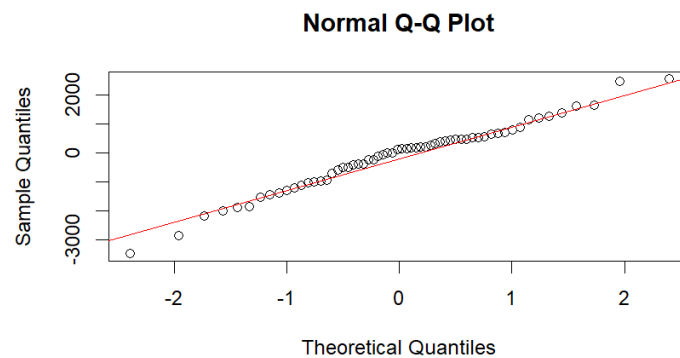


Figure 3.12: Normal Q-Q plot of residuals

From the Q-Q plot, most of the residuals are located on the straight line so the standard residuals of fitted model seems to be normal.

3.1.13 Histogram of Residuals

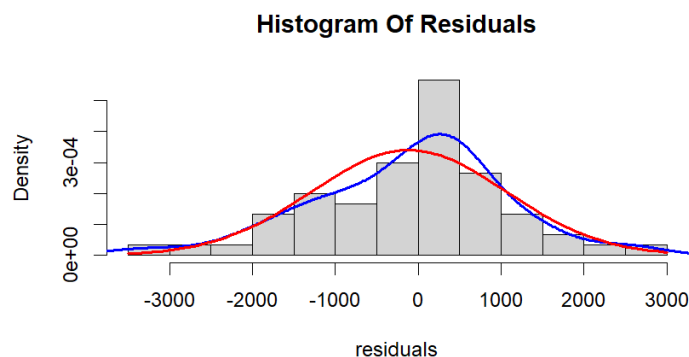


Figure 3.13: Normal Q-Q plot of residuals

The histogram of residuals indicates that the forecast model for COTPA Section 4 violations performs reasonably well. The residuals are approximately normally distributed, centering around zero, which suggests that the model predictions are unbiased on average. The bell-shaped curve of the histogram confirms that most residuals cluster close to zero, indicating a good fit.

Ljung-Box Test

To check the residuals from a time series model resemble white noise we use Ljung-Box test.

Here we test the hypothesis,

H_0 : The residuals are independently distributed (i.e., they resemble white noise).

Against

H_1 : The residuals are not independently distributed.

The result of the test is,

$Q^* = 4.682$, $df = 8$, $p\text{-value}=0.791$

Here p -value is 0.791 which is greater than 0.05 therefore we accept the null hypothesis that the data are independently distributed i.e, residuals are random.

Thus the fitted ARIMA model is valid.

Shapiro Wilk Test

Here we test the hypothesis,

H_0 : Residuals are from normal distribution.

Against

H_1 : Residuals are not from normal distribution.

The result of the test is,

$W = 0.97857$, $p\text{-value} = 0.3712$

Here p -value is greater than 0.05 therefor we accept the null hypothesis that the residual distribution is normal distribution.

3.1.14 forecasts from SARIMA

Forecast from Seasonal ARIMA(2, 1, 1)(1, 0, 0)_[12]

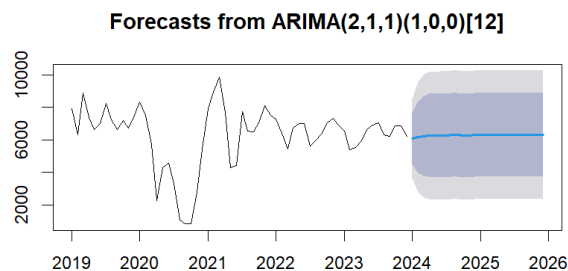


Figure 3.14: Forecasted plot

The table below shows the actual value and the forecasted value of COTPA Section 4 violation from dates 2023 January to 2023 December using SARIMA(2,1,1)(1,0,0)

2023	Actual values	Fitted values
January	6518	6594.913
February	5382	6470.561
March	5544	6450.701
April	5936	6300.566
May	6615	6259.256
June	6901	6254.063
July	7040	6369.256
August	6326	6338.469
September	6222	6310.592
October	6875	6257.016
November	6878	6233.622
December	6224	6267.845

Table 3.1: Actual vs Fitted values

The fitted values tend to slightly overestimate in January, February, and March. and the fitted values are closer to the actual values in the mid-year months (April to September). There is a difference in December, where the actual value is somewhat lower than the fitted value.

3.1.15 Forecast table of SARIMA(2, 1, 1)(1, 0, 0)_[12] for 2024

Month	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2024	6071.898	4547.537	7596.259	3740.589	8403.207
Feb 2024	6177.910	4035.420	8320.400	2901.255	9454.566
Mar 2024	6232.558	3843.759	8621.358	2579.205	9885.912
Apr 2024	6262.525	3793.558	8731.491	2486.566	10038.483
May 2024	6249.092	3758.482	8739.702	2440.032	10058.152
Jun 2024	6248.351	3752.896	8743.806	2431.882	10064.820
Jul 2024	6247.264	3751.122	8743.405	2429.745	10064.782
Aug 2024	6307.047	3811.161	8802.933	2489.919	10124.176
Sep 2024	6315.812	3820.323	8811.300	2499.291	10132.332
Oct 2024	6263.855	3768.689	8759.022	2447.827	10079.883
Nov 2024	6263.044	3768.081	8758.007	2447.328	10078.760
Dec 2024	6314.213	3819.357	8809.069	2498.661	10129.766

Table 3.2: Forecast Values for 2024

All values are within the control limits. Hence the Seasonal ARIMA model is appropriate for forecasting for the year 2024.

3.1.16 Forecast table of SARIMA(2, 1, 1)(1, 0, 0)_[12] for 2025

Month	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2025	6325.953	3742.901	8909.004	2375.517	10276.388
Feb 2025	6317.455	3735.318	8899.593	2368.417	10266.494
Mar 2025	6313.089	3732.133	8894.045	2365.857	10260.321
Apr 2025	6310.709	3730.712	8890.707	2364.944	10256.475
May 2025	6311.770	3732.313	8891.227	2366.831	10256.708
Jun 2025	6311.833	3732.592	8891.074	2367.225	10256.441
Jul 2025	6311.923	3732.727	8891.120	2367.382	10256.464
Aug 2025	6307.209	3727.991	8886.426	2362.636	10251.781
Sep 2025	6306.519	3727.268	8885.769	2361.896	10251.142
Oct 2025	6310.619	3731.343	8889.896	2365.956	10255.282
Nov 2025	6310.684	3731.391	8889.976	2365.996	10255.371
Dec 2025	6306.646	3727.345	8885.947	2361.946	10251.346

Table 3.3: Forecast Values for 2025

Here all values are within the control limits. Hence the SeasonalArima model is appropriate for forecasting for the year 2025.

Chapter 4

CONCLUSION

4.1 Conclusion

In this study, we used most violated section data of COTPA from January 2019 to December 2023 to forecast for the years 2024 and 2025.

For forecasting we used R programming to carry out

SeasonalARIMA(2, 1, 1)(1, 0, 0)_[12] model fitting. Based on the assumptions we checked whether the fitted models are perfect.

The forecast for 2024 regarding the most violated section of COTPA indicates varied trends throughout the year. The Seasonal ARIMA model projects periods of significant increase in violations, particularly in the middle of the year, suggesting a growing need for intensified enforcement and public awareness initiatives during these times. Other periods show stability or slight decreases in violations, which may indicate the effectiveness of ongoing preventive measures or enforcement consistency. This forecast highlights the importance of strategic planning in enforcement actions and educational campaigns.

The forecast for 2025 indicates occasional variations likely but not deviating significantly from the projected average. This stability can help authorities in planning enforcement strategies effectively throughout the year.

Chapter 5

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